

Evaluation of Weather Forecast with DL

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Abstract—In the last decade Weather forecasting has attained a lot of attention from all different groups and communities due to its vast effect on human life all around the globe. Widespread availability of weather observation data sets and the research and techniques available in deep learning sector have motivated many researchers and to explore hidden hierarchical patterns in vast quantities from the available data sets. The goal of meteorological scientists has always been to anticipate the weather accurately and in a timely manner. Traditional theory-driven numerical weather prediction systems face many challenges including a lack of understanding of physical causes, difficulties in extracting usable information from a flood of observation data, and the need for significant computational resources. The successful implementation of data-driven deep learning methods in a variety of domains, including computer vision, speech recognition, and time series prediction, has demonstrated that deep learning methods can efficiently extract temporal and spatial features from spatio-temporal data. [1] According to research, there is some evidence present which suggests that better weather forecasts can be produced by the introduction of big data mining and neural networks into the weather prediction workflow. [2] In this paper I shall be using different techniques through which I shall try to evaluate Weather Forecasting using Deep Learning techniques. The discussion will include an analysis of present Machine learning concepts and their applicability to the weather data with its statistical characteristics. Deep Learning based weather prediction is expected to be a strong supplement to the conventional method. In this paper I shall review the state of the art research on deep learning based weather forecasting, from the perspectives of Neural Network architecture design, geographical and temporal scales, data sets, and benchmarks. After that, I shall compare the benefits and drawbacks of Weather forecasting with Deep Learning to traditional Numerical Weather prediction and describe the potential future research subjects for DLWP.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Due to its impact on worldwide human life, weather forecasting has attracted the interest of numerous scholars from diverse research areas. The histories of numerical weather prediction (NWP) and machine learning (ML) or artificial intelligence (AI) (for the purposes of this research, the names are interchangeable) are vastly different. Lewis Fry Richardson explored manual NWP in Britain in 1922, and the first computer-aided weather forecasts were published in 1950 [3]. Operational weather forecasts in Sweden, the United States, and Japan followed soon after. While the evolution of NWP has been steady [4], the history of ML [5] has been more disruptive: McCulloch & Pitts proposed the first neural network (NN) in 1943 [6]. The subject developed until

the early 1960s, when existing algorithms were discovered to be inefficient and unstable. Backpropagation's development in 1970 [7], [8] ushered in a second wave of machine learning applications, as it became possible to build bigger NNs and train them to recognize nonlinear correlations in data [9]. Even though ML algorithms were still being developed, interest in them declined because they rarely showed significant performance gains and computing resources were insufficient to address larger issues. Large amounts of labelled data, which are required for most data-driven ML techniques, were also difficult to come by.

Weather forecasting is a vast field of research with wide potential applications which range from flight navigation, tropical cyclones, agriculture and travelling expeditions. Constraints of weather forecasting include learning of massive volume datasets and then developing an executable weather prediction model which revolves around learning structural patterns in that huge data set.

In recent times, a lot of research has happened in forecasting weather using statistical modeling techniques including machine learning which have provided successful results. [10]–[12]. Significant developments around 2010 kicked off the third wave of artificial intelligence, which is still going strong today: computing capabilities were vastly expanded thanks to massive parallel processing in graphical processing units (GPUs), convolutional neural networks (CNN) allowed for much more efficient analysis of massive (image) datasets, and large benchmark datasets were made available on the internet.

The studies in the fields of deep belief nets [13], deep networks [14], energy-based models [15] have acted as the base and foundation of deep learning approach as the deep architecture of generative models in the recent decade. The term "deep" in deep learning refers to a neural network (NN) that has more layers than "shallow" neural networks utilized in traditional machine learning models. After the successful execution of the layer-wise unsupervised pre-training method, which is used to efficiently handle the training issues, this multi-layer NN has sparked a lot of research interest. When compared to shallow models, the "deep" architecture is particularly important because NN with deep architecture can provide higher learning ability. [16]

Deep learning applications in various sectors have successfully demonstrated by many researchers, have motivated its utilization in weather forecasting and its modeling. A recent study [17], suggests to represent weather using hierarchical

feature that is obtained from a large number of weather data using Deep Neural Network(DNN).We take this opportunity to expand on the study of [18] and investigate the relevance of such methods to the NWP workflow as newer DL methods begin to supply such concepts.

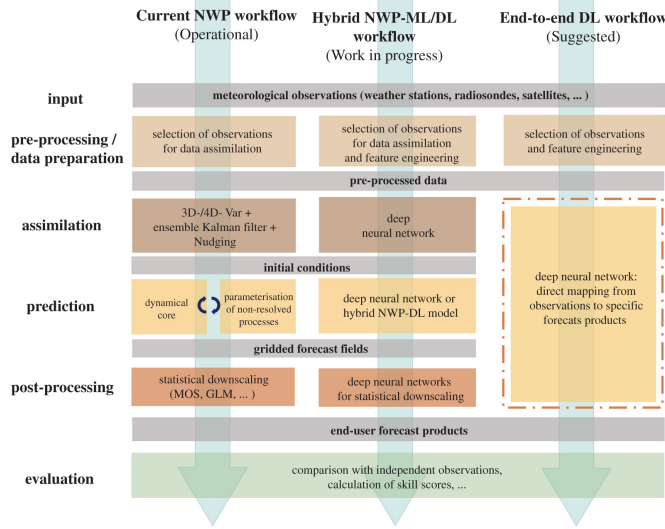


Fig. 1. Idealized workflows of current numerical weather prediction (left), next-generation weather prediction with individual components substituted or augmented by ML and DL techniques (centre), and a purely data-driven DL weather forecasting system (right). [2]

While much work is being done to replace particular sections of the NWP cycle with DL methods. In this paper I would try to investigate whether it is possible to replace all parts of the NWP workflow with one deep learning NN, which would take observations as input and directly produce end-user forecast solutions from the data as illustrated in Fig. 1.

Personal Motivation

The goal of my research is to look at the possibilities of deep learning techniques for weather forecasting using rich hierarchical weather representations learnt from large amounts of weather data. The models are tested using a large volume of weather data provided by BMKG (Indonesian Agency for Meteorology, Climatology, and Geophysics) collected from a number of weather stations in the Aceh area from 1973 to 2009, as well as the El-Nino Southern Oscillation (ENSO) data set provided by international institutions such as the National Weather Service Center for Environmental Prediction Climate Change (NOAA). Several deep learning methods for weather forecasting, such as the Conditional Restricted Boltzmann Machine (CRBM) and Convolutional Neural Network (CNN), will be investigated and compared with popular time series forecasting models, such as the Recurrent Neural Network (NN). [16]

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ACKNOWLEDGMENT

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